Executive Analytical Report: Collections Performance & Strategy Audit

This report audits 12 months of collections data across multiple operational channels to verify the reported 11% month-over-month (MoM) recovery improvement and determine the optimal allocation of a ₹10 Cr investment.

1. Data Quality & Forensics (The "Golden Dataset")

To create a reliable baseline from the 17 fragmented raw datasets, we resolved several critical, systemic data traps:

- **Phantom Recovery:** We identified duplicate payment events and duplicate payment references that falsely inflated gross recovery. These were strictly deduplicated.
- **Timezone Skew:** Event timestamps spanning UTC, Asia/Kolkata, and Asia/Dubai were standardized to UTC to protect driver analysis.
- **Identity Fragmentation:** We applied entity resolution to correct multiple agent identifiers and inconsistent campaign definitions.
- **Temporal Latency:** We isolated late-arriving events and treated status history as an append-only log anchored to actual event times rather than delayed system records.
- **Disposition Code Drift:** We harmonized changed schema versions and legacy disposition codes to fix the underreporting of successful conversion metrics.

2. Is the 11% MoM Improvement Real?

- The reported 11% MoM improvement is false.
- The figure was heavily skewed by the ingestion of exactly 500 duplicate payment retries that artificially inflated the raw recovery metrics by ₹2.59 Cr.
- After removing phantom recoveries and correcting metrics to include legacy disposition codes, true performance is statistically flat and highly volatile.

3. Drivers of Change & Counterfactual Analysis

Leadership introduced new targeting strategies (v1, v2, v3) midway through the year. To isolate their true impact against the legacy baseline, we applied a risk-stratified matching model.

Initially, the new strategies appeared highly successful, but this was a false signal caused by **attribution-window bias**. Treatment accounts simply had more days to accrue payments. To reveal the true ROI, we applied the following normalization formula:

True Incremental Lift = *(Avg. Recovery in exact 90-Days Post-Treatment) — (Avg. Recovery in exact 90-Days Post-Control)*

Metric	Naive Calculation (Biased)	Golden Model (Strict 90-Day Window)
Observation Window	Lifetime (Treatment > Control)	Exactly 90 Days post-intervention
Incremental Lift (per account)	+₹2,411.33	**+₹1,094.47**
Statistical Significance	Invalidated (Time Bias)	**Not Significant (CI: -₹329 to ₹2,304)**

4. Strategic Recommendation

- **Recommendation: HOLD the ₹10 Cr investment.** Do not immediately deploy capital into Better borrower targeting, telephony infrastructure, AI voice automation, or field operations.
- **Financial Justification:** As shown in the table above, when properly isolated for time biases, the incremental recovery per account shrinks to ₹1,094. Because the 95% Confidence Interval crosses zero, this lift is statistically indistinguishable from noise.
- **Next Steps:** The data is currently insufficient to make a reliable recommendation for full deployment. We recommend utilizing a fraction of the budget to initiate a strictly controlled A/B pilot with rigid 90-day observation windows to prove actual causal lift before committing the full ₹10 Cr.

5. Production Analytics Architecture

To provide daily, 60-second executive visibility, we recommend this data pipeline (Raw ➔ Staging ➔ Clean ➔ Golden ➔ Feature ➔ Metrics ➔ Dashboard):

- **Raw to Staging:** Ingest all 17 sources without schema enforcement, normalize to UTC, and flag late-arriving data.
- **Clean to Golden:** Apply strict primary key deduplication and implement Slowly Changing Dimensions (SCD) to prevent attribution window bias.
- **Feature to Metrics:** Pre-calculate complex metrics (e.g., True PTP Rate, Recovery per Account) using explicit data contracts.
- **Monitoring & Anomaly Detection:** Deploy automated daily checks to intercept duplicate payments or conflicting timestamps before they reach the dashboard.